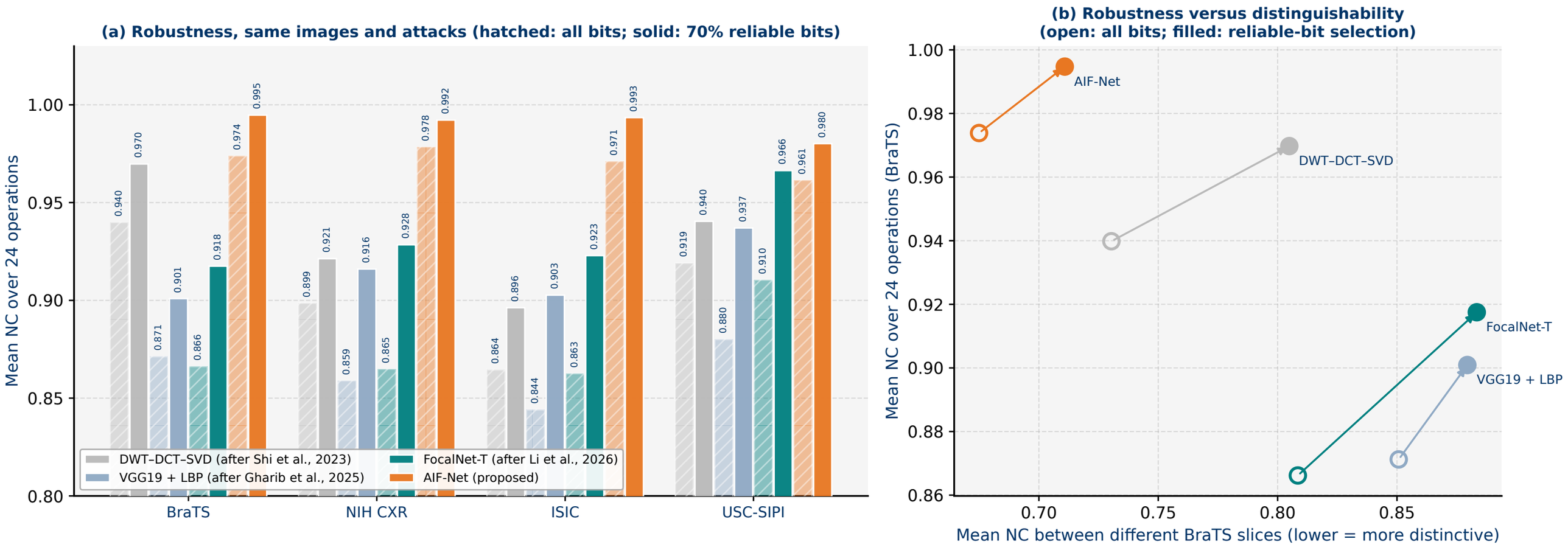
(a) Robustness, same images and attacks (hatched: all bits; solid: 70% reliable bits)
Mean NC over 24 operations
1.00
0.95
0.90
0.85
0.80
0.940
0.970
0.871
0.901
0.866
0.918
0.974
0.995
0.899
0.921
0.859
0.916
0.865
0.928
0.978
0.992
0.864
0.896
0.844
0.903
0.863
0.923
0.971
0.993
0.919
0.940
0.880
0.937
0.910
0.966
0.961
0.980
BraTS
NIH CXR
ISIC
USC-SIPI
DWT–DCT–SVD (after Shi et al., 2023)
VGG19 + LBP (after Gharib et al., 2025)
FocalNet-T (after Li et al., 2026)
AIF-Net (proposed)
(b) Robustness versus distinguishability
(open: all bits; filled: reliable-bit selection)
Mean NC over 24 operations (BraTS)
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.70
0.75
0.80
0.85
Mean NC between different BraTS slices (lower = more distinctive)
AIF-Net
DWT–DCT–SVD
FocalNet-T
VGG19 + LBP